

The objective of this project is to analyze superstore sales using EDA and find out the factors affecting sales.

```
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

1. LOADING DATA

```
df = pd.read_csv("superstore_final_dataset (1).csv", encoding='latin1')
```

```
df.head()
df.info()
df.describe()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 9800 entries, 0 to 9799
Data columns (total 18 columns):
#   Column                Non-Null Count  Dtype
---  -
0   Row_ID                 9800 non-null  int64
1   Order_ID               9800 non-null  object
2   Order_Date             9800 non-null  object
3   Ship_Date              9800 non-null  object
4   Ship_Mode              9800 non-null  object
5   Customer_ID            9800 non-null  object
6   Customer_Name          9800 non-null  object
7   Segment                9800 non-null  object
8   Country                9800 non-null  object
9   City                   9800 non-null  object
10  State                  9800 non-null  object
11  Postal_Code            9789 non-null  float64
12  Region                 9800 non-null  object
13  Product_ID             9800 non-null  object
14  Category               9800 non-null  object
15  Sub_Category           9800 non-null  object
16  Product_Name           9800 non-null  object
17  Sales                  9800 non-null  float64
dtypes: float64(2), int64(1), object(15)
memory usage: 1.3+ MB
```

	Row_ID	Postal_Code	Sales
count	9800.000000	9789.000000	9800.000000
mean	4900.500000	55273.322403	230.769059
std	2829.160653	32041.223413	626.651875
min	1.000000	1040.000000	0.444000
25%	2450.750000	23223.000000	17.248000
50%	4900.500000	58103.000000	54.490000
75%	7350.250000	90008.000000	210.605000
max	9800.000000	99301.000000	22638.480000

```
df.isnull().sum()
```

	0
Row_ID	0
Order_ID	0
Order_Date	0
Ship_Date	0
Ship_Mode	0
Customer_ID	0
Customer_Name	0
Segment	0
Country	0
City	0
State	0
Postal_Code	11
Region	0
Product_ID	0
Category	0
Sub_Category	0
Product_Name	0
Sales	0

dtype: int64

```
df.dropna(inplace=True)
```

```
df.drop_duplicates(inplace=True)
```

NO DUPLICATE DATA

```
df['Order_Date'] = pd.to_datetime(df['Order_Date'], dayfirst=True)
```

```
#YEAR
df['Year'] = df['Order_Date'].dt.year
#MONTH
df['Month'] = df['Order_Date'].dt.month
#MONTH NAME
df['Month_Name'] = df['Order_Date'].dt.month_name()
```

```
print(df.shape)
print(df.columns)
df.head()
```

```
(9789, 21)
Index(['Row_ID', 'Order_ID', 'Order_Date', 'Ship_Date', 'Ship_Mode',
      'Customer_ID', 'Customer_Name', 'Segment', 'Country', 'City', 'State',
      'Postal_Code', 'Region', 'Product_ID', 'Category', 'Sub_Category',
      'Product_Name', 'Sales', 'Year', 'Month', 'Month_Name'],
      dtype='object')
```

	Row_ID	Order_ID	Order_Date	Ship_Date	Ship_Mode	Customer_ID	Customer_
0	1	CA-2017-152156	2017-11-08	11/11/2017	Second Class	CG-12520	Claire
1	2	CA-2017-152156	2017-11-08	11/11/2017	Second Class	CG-12520	Claire
2	3	CA-2017-138688	2017-06-12	16/06/2017	Second Class	DV-13045	Darrin Van
3	4	US-2016-108966	2016-10-11	18/10/2016	Standard Class	SO-20335	Sean O Do
4	5	US-2016-108966	2016-10-11	18/10/2016	Standard Class	SO-20335	Sean O Do

5 rows × 21 columns

```
print(df.columns.tolist())
```

```
['Row_ID', 'Order_ID', 'Order_Date', 'Ship_Date', 'Ship_Mode', 'Customer_ID',
```

2. CREATING THE KPI's

```
total_revenue = df['Sales'].sum()

print("Total Revenue:", total_revenue)
```

Total Revenue: 2252607.4127

```
total_orders = df['Order_ID'].nunique()
```

```
print("Total Orders:", total_orders)
```

```
Total Orders: 4916
```

```
total_customers = df['Customer_ID'].nunique()
```

```
print("Total Customers:", total_customers)
```

```
Total Customers: 793
```

```
aov = total_revenue / total_orders
```

```
print("Average Order Value:", round(aov, 2))
```

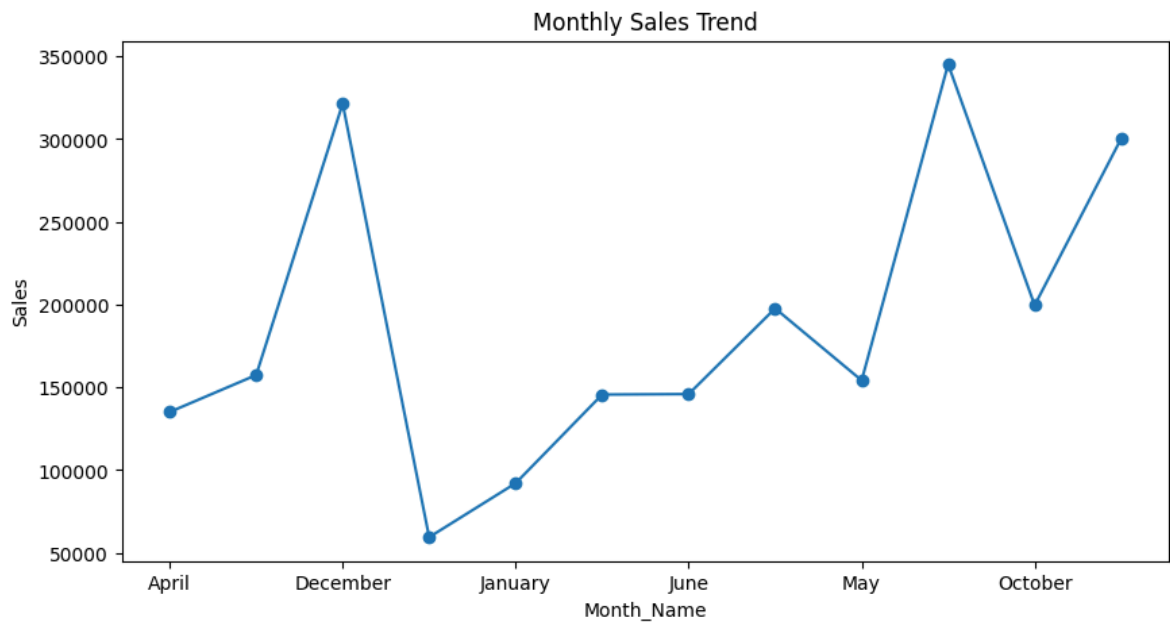
```
Average Order Value: 458.22
```

3. monthly sales trend

```
monthly_sales = df.groupby('Month_Name')['Sales'].sum()
```

```
monthly_sales.plot(  
    kind='line',  
    marker='o',  
    figsize=(10,5)  
)
```

```
plt.title("Monthly Sales Trend")  
plt.ylabel("Sales")  
plt.show()
```

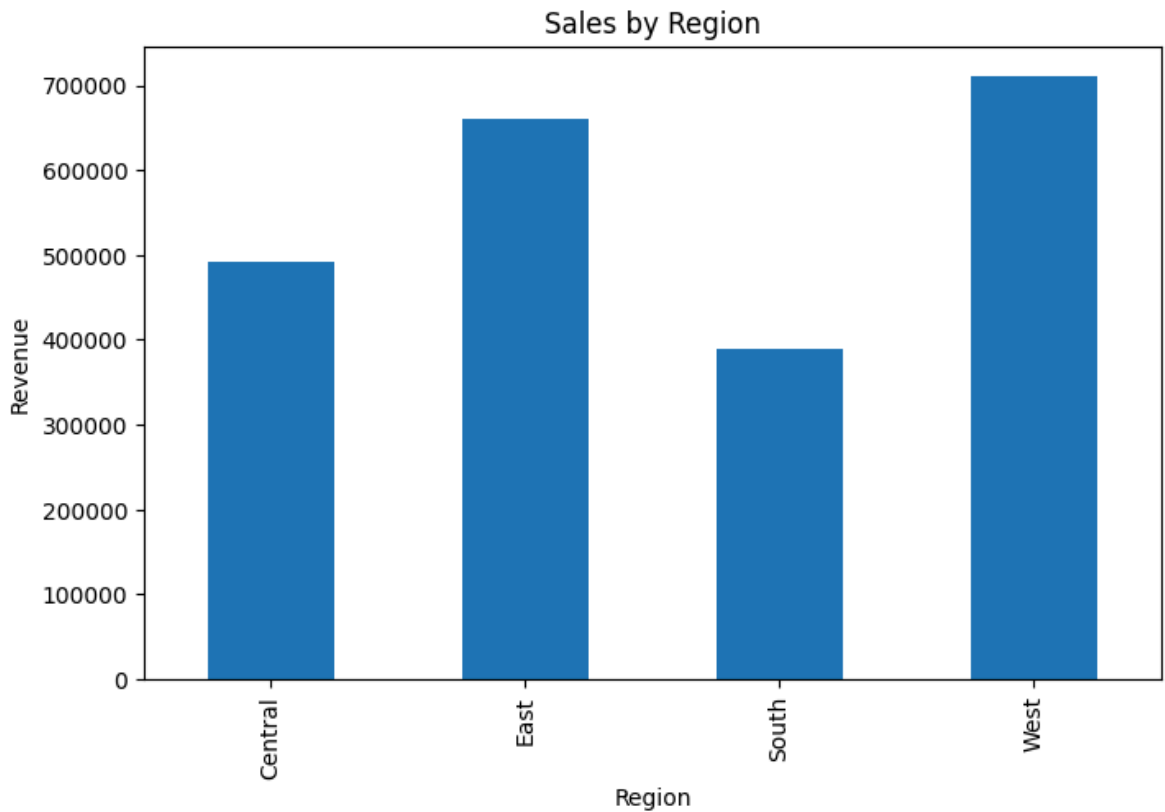


4.regional analysis

```
region_sales = df.groupby('Region')['Sales'].sum()

region_sales.plot(
    kind='bar',
    figsize=(8,5)
)

plt.title("Sales by Region")
plt.ylabel("Revenue")
plt.show()
```

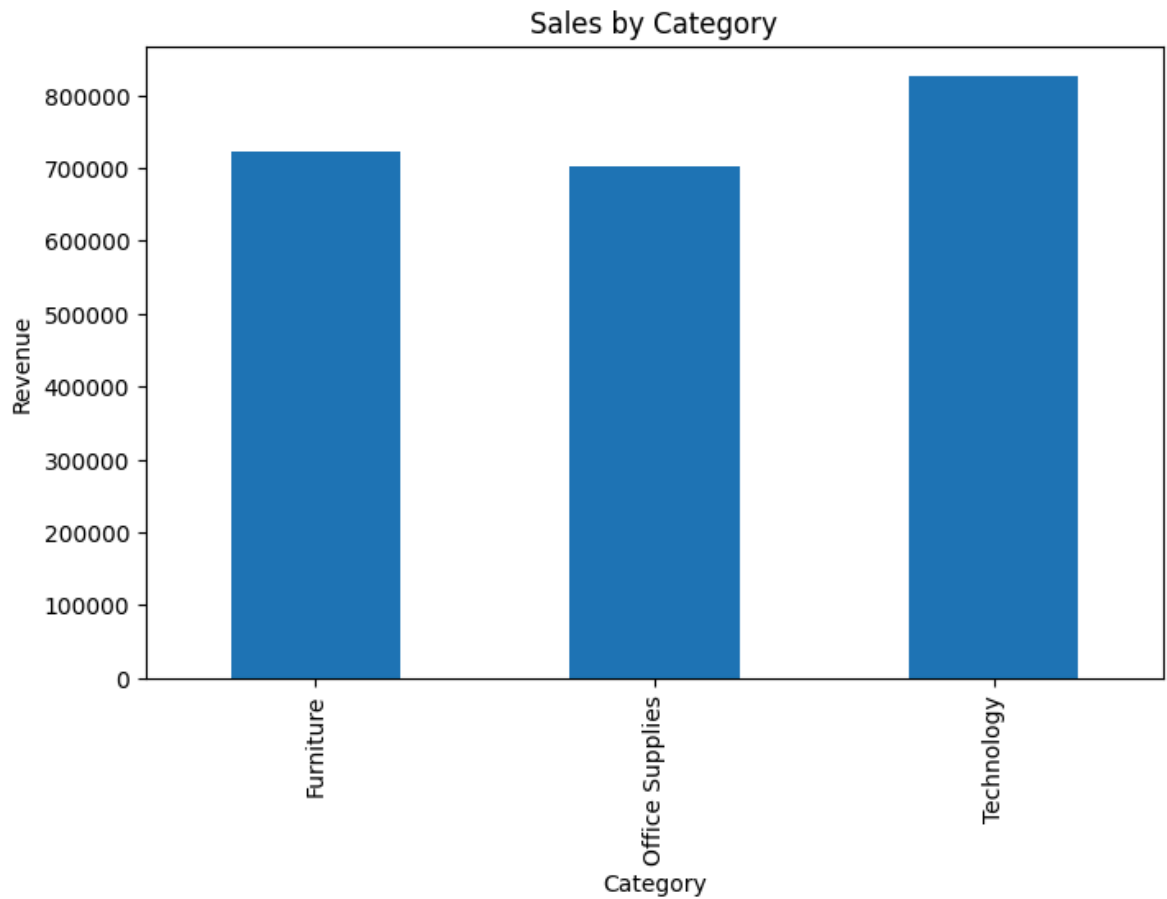


5. catogory analysis

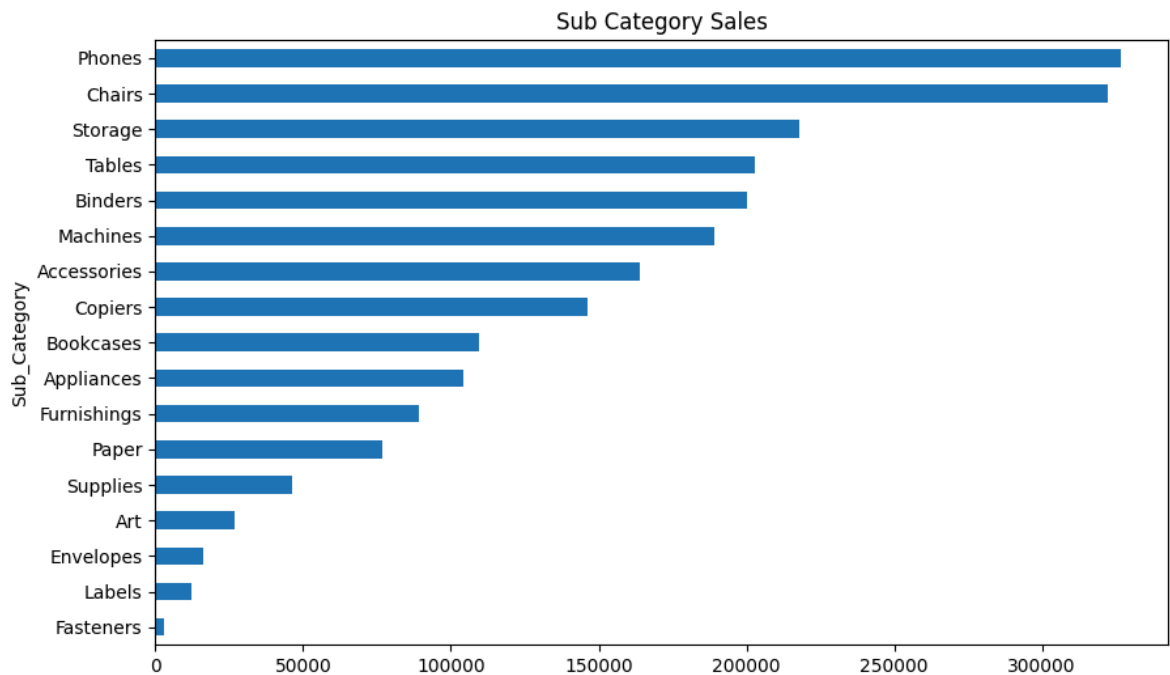
```
category_sales = df.groupby('Category')['Sales'].sum()

category_sales.plot(
    kind='bar',
    figsize=(8,5)
)

plt.title("Sales by Category")
plt.ylabel("Revenue")
plt.show()
```



```
subcategory_sales = df.groupby(  
    'Sub_Category'  
)['Sales'].sum()  
  
subcategory_sales.sort_values().plot(  
    kind='barh',  
    figsize=(10,6)  
)  
  
plt.title("Sub Category Sales")  
plt.show()
```

6. top 10 products and its visualization

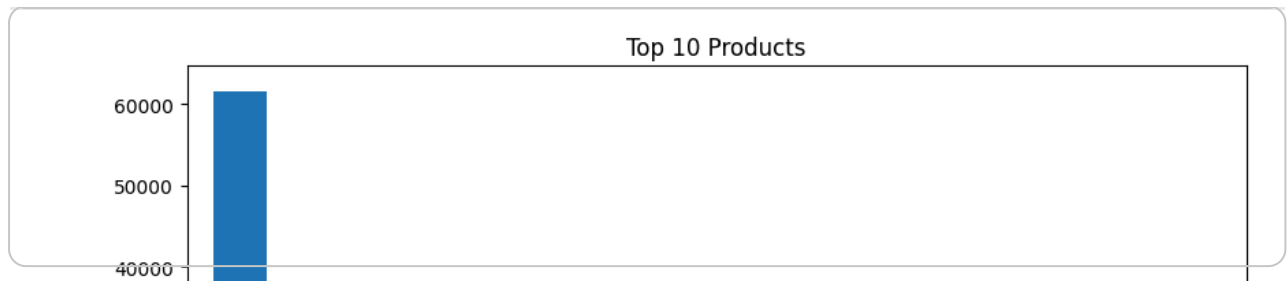
```
top_products = df.groupby(
    'Product_Name'
)['Sales'].sum()

top10 = top_products.sort_values(
    ascending=False
).head(10)

print(top10)
```

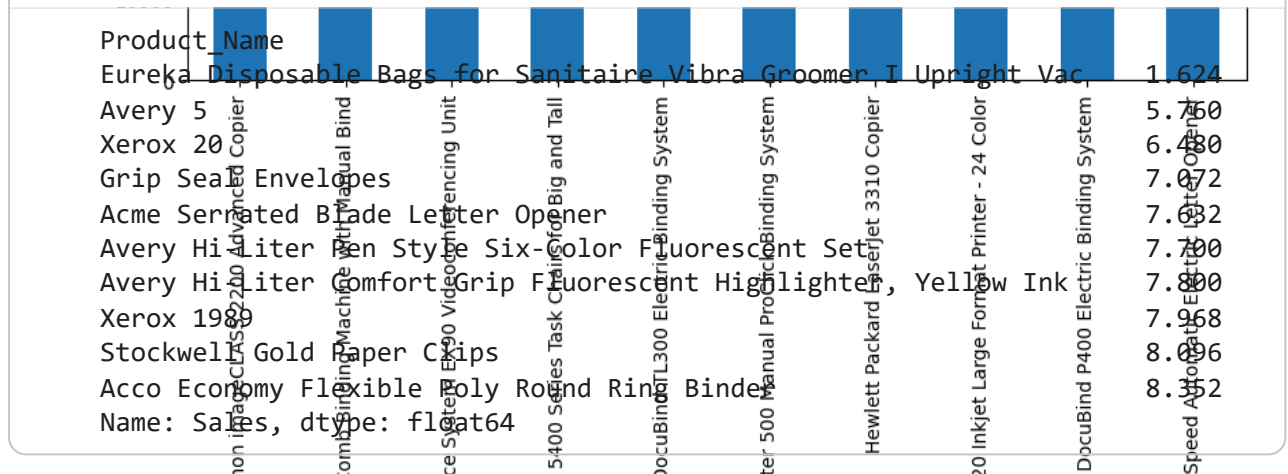
```
Product_Name
Canon imageCLASS 2200 Advanced Copier
Fellowes PB500 Electric Punch Plastic Comb Binding Machine with Manual Bind
Cisco TelePresence System EX90 Videoconferencing Unit
HON 5400 Series Task Chairs for Big and Tall
GBC DocuBind TL300 Electric Binding System
GBC Ibimaster 500 Manual ProClick Binding System
Hewlett Packard LaserJet 3310 Copier
HP Designjet T520 Inkjet Large Format Printer - 24 Color
GBC DocuBind P400 Electric Binding System
High Speed Automatic Electric Letter Opener
Name: Sales, dtype: float64
```

```
top10.plot(  
    kind='bar',  
    figsize=(10,5)  
)  
  
plt.title("Top 10 Products")  
plt.show()
```



```
bottom10 = df.groupby(
    'Product_Name'
)['Sales'].sum().sort_values().head(10)

print(bottom10)
```



```
segment_sales = df.groupby(
    'Segment'
)['Sales'].sum()

segment_sales.plot(
    kind='pie',
    autopct='%1.1f%%'
)

plt.title("Sales by Customer Segment")
plt.show()
```

